HUMAN ACTIVITY RECOGNITION DATA ANALYSIS

Introduction

This Project analyses Human Activity Recognition Dataset, which consists of sensor signals that are collected from experiments with 30 volunteers conducting 6 daily activities (Walking, Walking Upstairs, Walking Downstairs, Sitting, Standing, Laying) while wiring Samsung Galaxy S II smartphone on the waist.

The dataset contains 33 features in which 20 features are Time domain features and 13 features are Frequency domain features derived via Fast Fourier Transform (FFT). This dataset is a simplified version of a dataset from UCI Machine Learning Repository which originally contains 561 features.

The primary objective of this report is to analyse the latent underlying structures of this dataset using multivariate methods. Whether some signals are redundant. We examine whether the observed variables can be explained by a smaller number of underlying factors or independent components, and how different methodologies will capture this structure. We also assess the validity of model assumptions, compare methodological outcomes, and interpret the resulting latent dimensions in the context of human physical activity.

Data Inspection:

This dataset has no missing values. Outliers were identified using a z-score threshold of 5, which is a conservative criterion designed to detect only highly unusual observations.

Overall, the number of detected outliers is very small relative to the total sample size (10,299 observations), with most features exhibiting fewer than 50 extreme values. This corresponds to a negligible proportion of the data (well below 1%), indicating that the dataset is largely well-behaved and free from substantial anomalies.

Outliers are primarily concentrated in certain raw sensor features, particularly the tri-axial body acceleration and gyroscope measurements (e.g., tBodyAcc-mean()-Z, tBodyAcc-mean()-Y, and tBodyGyro-mean()-Z). In contrast, magnitude-based and frequency-domain features exhibit almost no extreme values.

This pattern suggests that directional signals (X, Y, Z components) are more susceptible to variability, likely reflecting genuine differences in movement intensity across activities, especially along the vertical axis. Meanwhile, aggregated features such as magnitudes and frequency summaries appear more stable due to their smoothing and transformation properties.

Due to the low proportion of outliers and retaining these outliers preserves the natural variability of human motion, which is important for meaningful latent structure analysis, no outliers were removed.

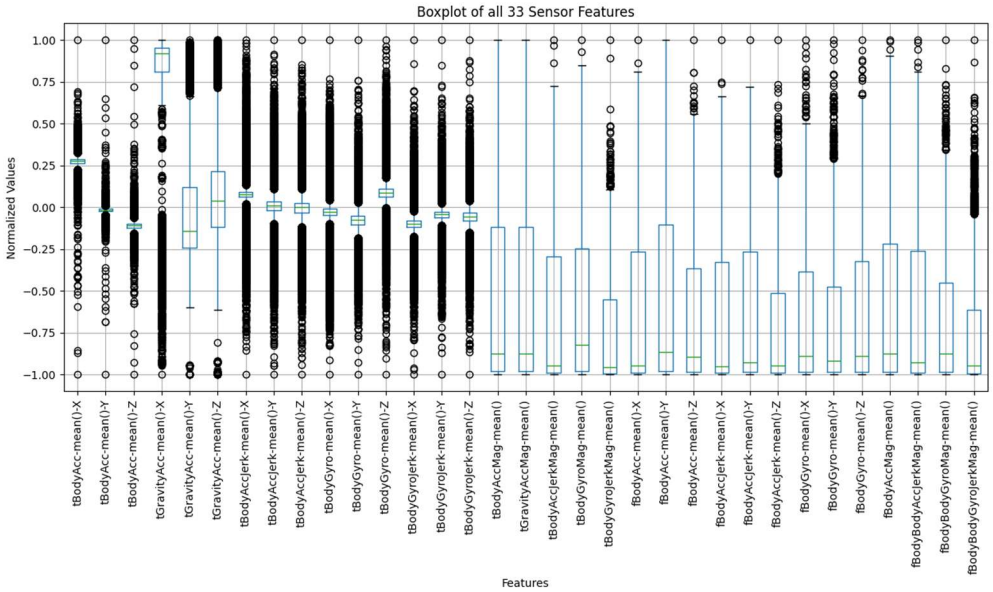

Figure 1. Boxplot of all 33 normalized sensor features with values scaled to [-1, 1]. The plot shows that frequency-domain features and time-domain features tend to exhibit relatively stable distributions, while several 3-Axial time-domain signals show greater variability and more extreme values. This indicates the presence of outliers primarily in directional time domain sensor measurements, suggesting heterogeneous variability across different types of features.

The boxplot (Figure 1) shows that frequency-domain features and the magnitude features from the time domain tend to have values centered around zero, including both positive and negative values. All 33 features are normalized and bounded within $[-1,1]$, but their variability differs substantially across features. Some features such as tBodyAcc-mean()-XYZ, tBodyAccJerk-mean()-XYZ, and tBodyGyro-mean()-XYZ exhibit relatively small variance based on the spread of the interquartile range (IQR), while other features such as fBodyAcc-mean()-XYZ show larger variance.

Given that these features are measured on a comparable normalized scale but still exhibit different variances, it is important to ensure that no subset of variables dominates the multivariate analysis purely due to scale differences. Therefore, we standardize the data prior to applying multivariate methods, so that each feature contributes more equally to the analysis.

Outliers are observed in some features, especially among the time-domain 3-axial signals. This is reasonable since abrupt human movements can generate extreme values. In contrast, magnitude-based features show fewer outliers, indicating more stable and aggregated behavior compared to axis-specific measurements.

The presence of outliers and the differences in variability across features suggest that the underlying distributions may deviate from normality and may exhibit heavier tails in some cases. Since methods such as PCA and Factor Analysis rely on Covariance matrix and are sensitive to the noise, these characteristics may affect their performance and interpretation. This motivates the need to assess distributional assumptions and interpret results with appropriate caution.

Correlation Matrix, Assumption Testing and Method Selection:

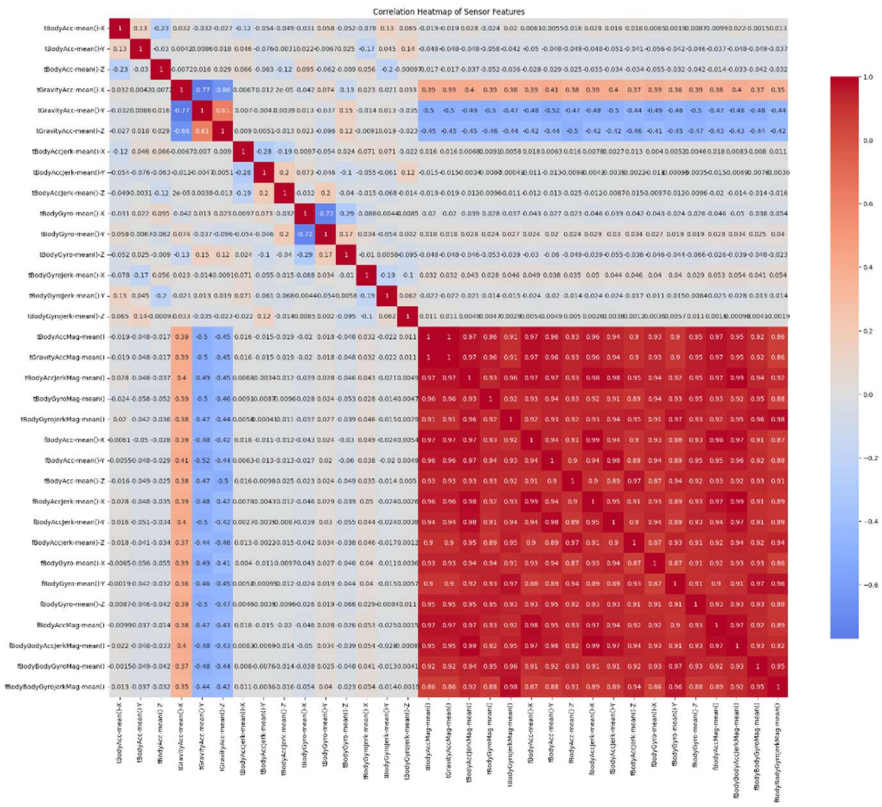

Figure 2. Correlation heatmap of the 33 features. The lower-right block shows strong correlations (coefficients > 0.8) among frequency-domain features and 5 magnitude signals time-domain features, indicating shared underlying structure. In contrast, 3-Axial time-domain features exhibit weak correlations with other variables, suggesting partial independence and supporting the existence of multiple latent structures rather than a single homogeneous feature group.

The correlation heatmap (Figure 2) shows that, in the lower-right block, the 13 frequency-domain features and the 5 magnitude time-domain features are highly correlated, with correlation coefficients generally larger than 0.8. This indicates the presence of strong shared variation among these variables, suggesting that they may be driven by a smaller number of underlying latent factors. Therefore, it is reasonable to consider Factor Analysis to identify such latent structures. In addition, the high correlations among these features suggest that the data may lie in a lower-dimensional subspace, making Principal Component Analysis (PCA) an appropriate baseline method for dimensionality reduction.

It is also noticeable that in the upper-right and lower-left blocks of the heatmap, most correlation values are close to zero. This indicates that the 3-Axial signals from the 20 time domain features are weakly correlated with the frequency-domain and magnitude features. As a result, methods that rely on relationships between two blocks of variables, such as Canonical Correlation Analysis (CCA) and Multivariate Linear Regression (MLR), are not particularly suitable for this dataset, since there is no clear pair of strongly related variable groups. However, this relative independence across groups of variables suggests that Independent Component Analysis (ICA) may be useful for extracting components that are statistically independent, potentially capturing distinct underlying signal sources.

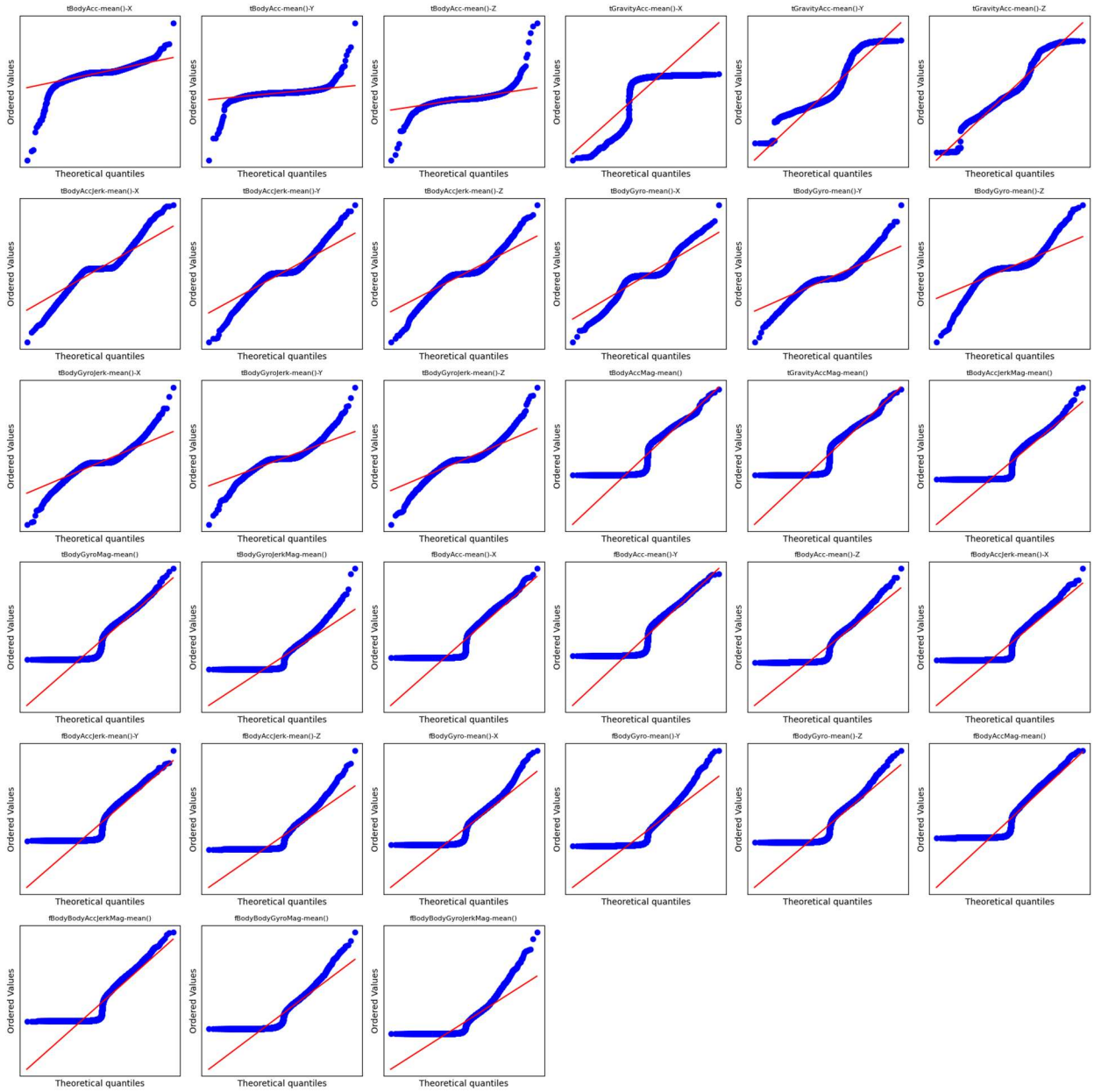

Figure 3. QQ plots comparing selected features against the theoretical normal distribution. Most plots deviate from the reference line, indicating that the variables do not follow univariate normal

distributions. Together with Mardia's test results, this provides evidence that the dataset does not satisfy the assumption of multivariate normality.

From the QQ plots above (Figure 3), we can see that most plots do not show a strong linear pattern, suggesting that most variables do not follow a univariate normal distribution. In addition, by conducting Mardia's test, we obtain Mardia's Skewness (b1p) = 623.4089416753558 with p-value = 0.0, and Mardia's Kurtosis (b2p) = 3244.0723845915772 with p-value = 0.0. Since both p-values are less than 0.05, we reject the null hypothesis that the dataset follows a multivariate normal distribution. Thus, we conclude that the dataset does not satisfy the assumption of multivariate normality. This is important because methods such as Factor Analysis often rely to varying degrees on normality assumptions, while Independent Component Analysis (ICA) explicitly benefits from non-Gaussianity. In particular, ICA assumes that the underlying sources are non-Gaussian and statistically independent, with at most one component being Gaussian. Therefore, the observed departure from multivariate normality provides justification for applying ICA to this dataset.

Model Specification and Interpretation:

Based on the observations above, several features are highly correlated, indicating redundancy among variables. Therefore, Principal Component Analysis (PCA) is applied to reduce the dimensionality of the dataset while retaining most of the variability. PCA achieves this by constructing orthogonal principal components that capture maximal variance.

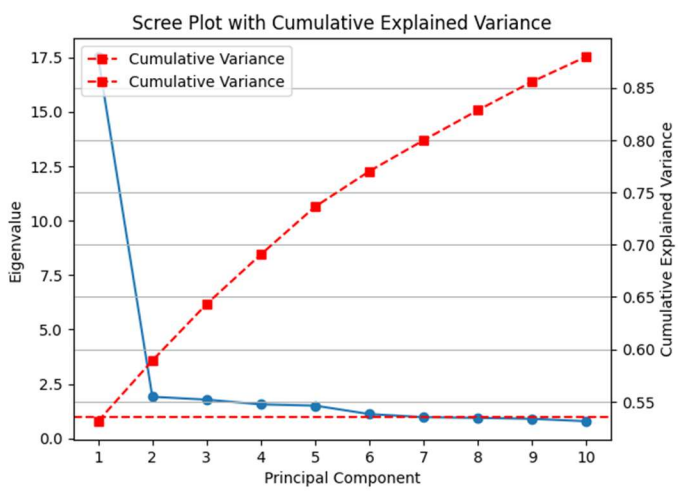

Figure 4. Scree plot of PCA eigenvalues. A clear elbow is observed around the first two principal components, with the first component explaining approximately 50% of the total variance. Based on the Kaiser criterion and cumulative variance, retaining approximately 6–7 principal components captures about 75%–80% of the total variability, suggesting effective dimensionality reduction from 33 variables to a much smaller set.

We conducted PCA and used a scree plot to help determine the number of components. The scree plot (Figure 4) shows a clear elbow around PC1 and PC2, with PC1 alone explaining approximately 50% of the total variance. According to the Kaiser criterion, about the first 6 principal components have eigenvalues greater than 1. Considering multiple criteria, including the eigenvalue ratios (>2) and cumulative variance, the first 6 principal components explain about 75% of the total variance, and the first 7 components explain about 80%.

Thus, PCA suggests that the original 33-dimensional dataset can be effectively reduced to approximately 6–7 dimensions while still preserving most of the variability in the data.

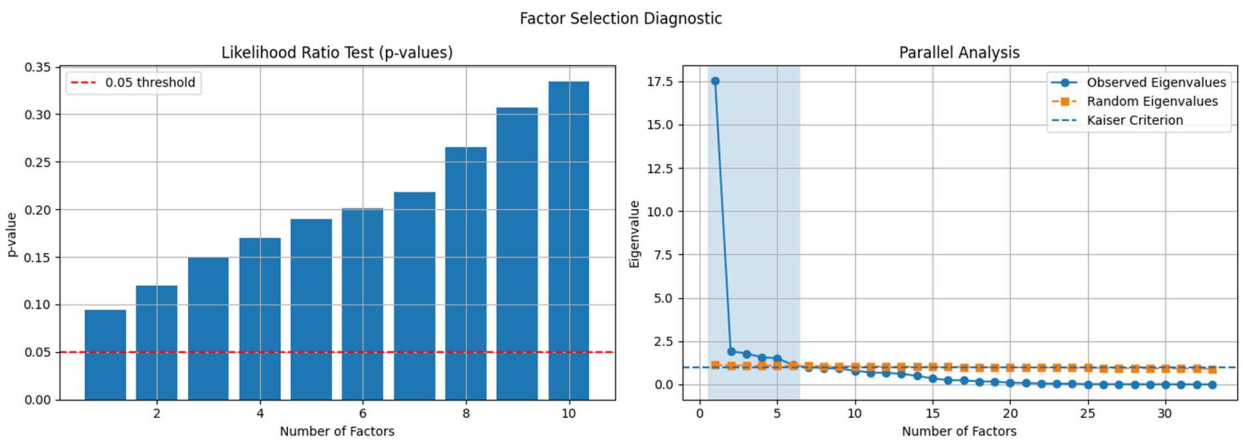

Figure 5. Results of parallel analysis and likelihood ratio test for factor selection. The parallel analysis indicates that the first 6 factors have observed eigenvalues larger than those generated from random data, suggesting that 6 factors are appropriate. The likelihood ratio test further supports this choice, as the model with 6 factors yields a non-significant p-value ($p > 0.05$), indicating adequate model fit compared to the saturated model.

Despite the dataset doesn't strictly follow a multivariate normal distribution, since we are interested in the underlying highly correlated features, we proceed with Factor Analysis (FA) to investigate latent factors, particularly among frequency-domain features and the 5 magnitude signals features. FA is appropriate here because it aims to model the covariance structure in terms of a smaller number of latent factors.

We first evaluate the suitability of FA using the KMO statistic and obtain a value of 0.9291, which is considered excellent and indicates strong shared correlation among variables. Bartlett's Sphericity Test could not produce a valid p-value due to numerical instability; however, given the high KMO value, there is sufficient evidence of correlation structure to justify factor extraction.

Based on the PCA results, we initially consider retaining 6–7 factors. To further validate this choice, we conducted parallel analysis and a likelihood ratio test. The parallel analysis (Figure 5) shows that the first 6 observed eigenvalues exceed those from randomly

generated data, suggesting that retaining 6 factors is appropriate. This aligns with the PCA results. The likelihood ratio test indicates that when the number of factors is 6, the p-value is greater than 0.05, meaning we fail to reject the null hypothesis that the 6-factor model fits adequately compared to the saturated model. Therefore, selecting 6 factors is a reasonable and supported choice.

Canonical Correlation Analysis (CCA) is designed to model relationships between two distinct sets of variables. Although the dataset contains both time-domain and frequency-domain features, the correlation structure does not support a clean separation into two independent blocks. 5 Magnitude signals of time-domain features exhibit strong correlations with frequency-domain features, while 3-Axial features (X, Y, Z) show weak correlations with other variables. This indicates that the features represent overlapping transformations of the same underlying signals rather than two clearly separable groups. Therefore, applying CCA would require an arbitrary partition of variables without a strong theoretical justification, and is unlikely to yield meaningful canonical relationships. For this reason, CCA is not pursued in this analysis. A similar argument applies to Multivariate Linear Regression (MLR), as there is no natural choice of predictors and responses among these variables.

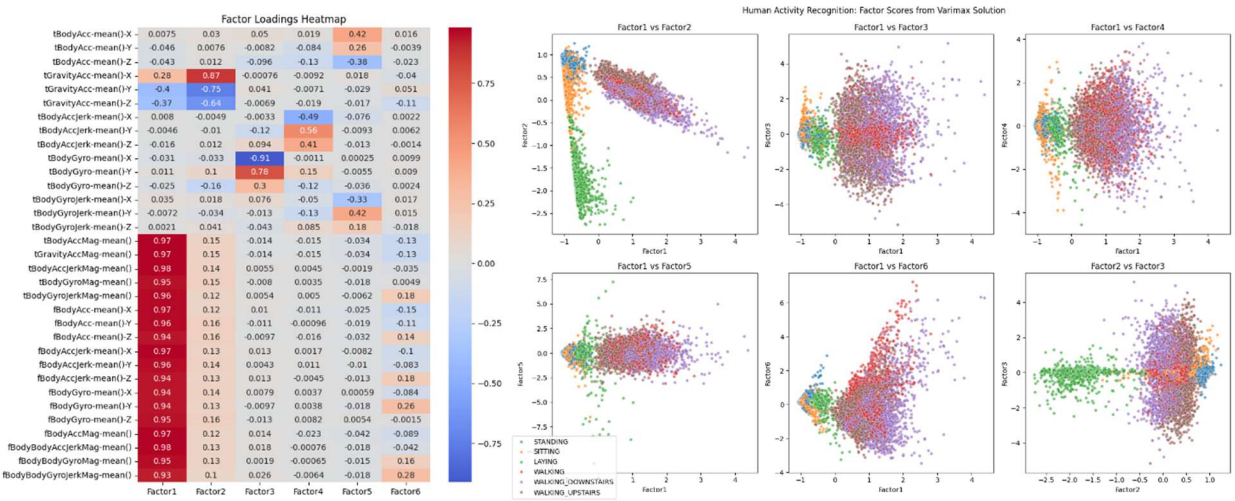

Figure 6. Factor loadings heatmap and factor score projections. The loadings show that Factor 1 is dominated by frequency-domain and 5 magnitude signals in time-domain features, while Factors 2–4 are associated with gravity-related signals and gyroscope measurements. Factor score plots indicate that Factor 1 effectively separates dynamic activities from static ones, while Factors 2 and 3 further distinguish specific activities such as lying. Overall, the factors correspond to meaningful physical interpretations of human motion, although higher-order factors exhibit weaker and less distinct loading patterns.

From the Factor Loadings Heatmap (Figure 6), Factor 1 is primarily dominated by frequency-domain features and 5 magnitude signals in time-domain features. Factor 2 is mainly associated with mean gravity acceleration features (tGravityAcc-mean()-XYZ).

Factor 3 is primarily related to mean body angular velocity (tBodyGyro-mean()-XYZ), while Factor 4 is associated with mean body angular velocity jerk (tBodyGyroJerk-mean()-XYZ). Factor 5 is linked to angular velocity jerk magnitude features (tBodyGyroJerkMag). Factor 6 does not exhibit a clear dominant pattern in the heatmap but appears to be more weakly associated with frequency-domain features.

The factor score plots (Figure 6) show that Factor 1 effectively separates dynamic activities (Walking, Walking Upstairs, Walking Downstairs) from static activities (Laying, Sitting, Standing). Factors 2 and 3 further help distinguish Laying from other activities.

Overall, the factor loadings reveal meaningful groupings of variables according to their physical interpretations, including magnitude-based signals, gravity-related components, and rotational motion features. Higher-order factors (Factors 4–6) exhibit weaker and less distinct loading structures, likely capturing more subtle variation or noise in the data.

As for ICA, the weak cross-correlation between time-domain axis-specific features and other feature groups, together with the evidence of non-Gaussianity from earlier tests, suggests that ICA may be suitable for extracting statistically independent components. However, due to visualization constraints (limited number of figures allowed), ICA results are not further explored in this report.

Discussion:

Overall, the results from PCA, FA, and ICA suggest that the HAR dataset exhibits a mixed structure consisting of both shared variation and partially independent components. On one hand, the strong correlations among frequency-domain features and magnitude signals in time domain features indicate the presence of underlying latent factors that capture common patterns of movement intensity. This is reflected in PCA, where a relatively small number of principal components explain a large proportion of the total variance, and in FA, where a limited number of factors successfully capture the shared covariance structure.

On the other hand, the weak correlations observed among axis-specific time-domain features suggest that not all variables are driven by the same underlying factors. This partial independence supports the consideration of ICA, which is designed to identify statistically independent sources. Although ICA is not fully explored in this report, its applicability is consistent with the non-Gaussian nature of the data and the weak dependence structure across certain feature groups.

The comparison between methods also highlights their different modeling assumptions. PCA focuses on variance maximization without distinguishing between shared and unique variance, while FA explicitly models latent variables that explain the correlations among observed features. ICA, in contrast, goes beyond Covariance matrix and aims to recover independent sources. The fact that PCA and FA produce consistent conclusions regarding

the number of latent dimensions further supports the robustness of the underlying low-dimensional structure.

At the same time, the results suggest that the feature space is not naturally partitioned into two independent blocks, which explains why methods such as CCA are not suitable in this context. Instead, the dataset appears to consist of overlapping representations of the same underlying physical processes, rather than clearly separable variable groups.

A limitation of this analysis is that the interpretation of latent factors remains partially subjective, particularly for higher-order factors that do not exhibit clear loading patterns. Additionally, the choice of the number of factors, although supported by multiple criteria, still involves some level of judgment. Future work could involve validating these latent structures using external labels or downstream prediction tasks to further assess their practical relevance.

AI Attribution and Acknowledgements

Generative AI (ChatGPT) was used for proofreading and improving the clarity of the text. The author independently conducted all analyses, made all modeling decisions, and verified all results.